```
from google.colab import files

uploaded = files.upload()

import pandas as pd

df = pd.read_csv("customers.csv")

df.head()
```

Choose Files customers.csv

customers.csv(text/csv) - 8495 bytes, last modified: 8/17/2026 - 100% done

Saving customers.csv to customers (1).csv

	customer_id	full_name	gender	birth_date	city	registration_date
0	1	Aylin İbrahimov	Qadın	1985-04-05	Gəncə	2025-11-23
1	2	Zəhra Məmmədov	Qadın	1978-02-07	Bakı	2025-09-08
2	3	Aysel Həsənov	Qadın	1985-08-19	Bakı	2024-01-07
3	4	Zəhra Rzayev	Qadın	1986-03-07	Lənkəran	2024-12-10
4	5	Nigar Həsənov	Qadın	1981-06-28	Bakı	2024-09-27

Next steps: Generate code with df | New interactive sheet

```
from google.colab import files

uploaded = files.upload()
df1 = pd.read_csv("orders.csv")

df1.head()
```

Choose Files orders.csv

orders.csv(text/csv) - 25355 bytes, last modified: 8/17/2026 - 100% done

Saving orders.csv to orders (2).csv

	order_id	customer_id	order_date	status	payment_method
0	1	55	2026-04-11	Completed	Kart
1	2	142	2025-07-30	Completed	Kart
2	3	92	2025-01-27	Completed	Kart
3	4	17	2026-03-08	Completed	Nağd
4	5	8	2026-06-18	Completed	Kart

Next steps: Generate code with df1 | New interactive sheet

```
from google.colab import files

uploaded = files.upload()
df2 = pd.read_csv("products.csv")

df2.head()
```

Choose Files products.csv

products.csv(text/csv) - 816 bytes, last modified: 8/17/2026 - 100% done

Saving products.csv to products (2).csv

	product_id	product_name	category	brand	unit_price	cost_price
0	1	Bluetooth qulaqcıq	Elektronika	Nova	59	32
1	2	Powerbank	Elektronika	Apex	74	43
2	3	Smart saat	Elektronika	Prime	129	78
3	4	Simsiz mouse	Elektronika	Nova	39	20
4	5	Elektrik çaydanı	Ev və Mətbəx	Luma	69	40

Next steps: Generate code with df2 New interactive sheet

```
from google.colab import files

uploaded = files.upload()

df3 = pd.read_csv("order_items.csv")
df3.head()
```

Choose Files order_items.csv

order_items.csv(text/csv) - 23828 bytes, last modified: 8/17/2026 - 100% done

Saving order_items.csv to order_items (1).csv

	order_item_id	order_id	product_id	quantity	unit_price	discount_percent
0	1	1	11	1	89	0
1	2	2	11	1	89	20
2	3	2	5	3	69	10
3	4	2	1	2	59	0
4	5	3	6	1	89	0

Next steps: Generate code with df3 New interactive sheet

```
##Null və duplicate yoxlanışı
datasets = {
    "customers": df,
    "orders": df2,
    "order_items": df3,
    "products": df4
}

for name, data in datasets.items():
    print("\n", "="*40)
    print(name)

    print("\nNull məlumatların sayı:")
    print(data.isnull().sum())

    print("\nDuplicate sətirlərin sayı:")
    print(data.duplicated().sum())
```

```
 ========================================
customers

Null məlumatların sayı:
customer_id          0
full_name            0
gender               0
birth_date           0
city                 0
registration_date    0
dtype: int64

Duplicate sətirlərin sayı:
0

 ========================================
orders

Null məlumatların sayı:
product_id      0
product_name    0
category        0
brand           0
unit_price      0
cost_price      0
dtype: int64

Duplicate sətirlərin sayı:
0

 ========================================
order_items
```

```
Null məlumatların sayı:
order_item_id       0
order_id            0
product_id          0
quantity            0
unit_price          0
discount_percent    0
dtype: int64

Duplicate sətirlərin sayı:
0

 ========================================
products

Null məlumatların sayı:
product_id      0
product_name    0
category        0
brand           0
unit_price      0
cost_price      0
dtype: int64
```

```
##sale_amount hesablanması
df3["sale_amount"] = (
    df3["quantity"] *
    df3["unit_price"] *
    (1 - df3["discount_percent"] / 100)
)

df3.head()
```

	order_item_id	order_id	product_id	quantity	unit_price	discount_percent	sale_amount
0	1	1	11	1	89	0	89.0
1	2	2	11	1	89	20	71.2
2	3	2	5	3	69	10	186.3
3	4	2	1	2	59	0	118.0
4	5	3	6	1	89	0	89.0

Next steps: Generate code with df3 New interactive sheet

```
print("Ümumi satış:", df3["sale_amount"].sum())
```

```
Ümumi satış: 93692.4
```

```
print(df3[[
    "quantity",
    "unit_price",
    "discount_percent",
    "sale_amount"
]].head(10))
```

```
   quantity  unit_price  discount_percent  sale_amount
0         1          89                 0        89.00
1         1          89                20        71.20
2         3          69                10       186.30
3         2          59                 0       118.00
4         1          89                 0        89.00
5         1          39                20        31.20
6         1          27                15        22.95
7         1          39                 5        37.05
8         2          68                 0       136.00
9         1          34                10        30.60
```

```
sales["sale_amount"] = (
    sales["quantity"] *
    sales["unit_price_x"] *
    (1 - sales["discount_percent"] / 100)
)

display(sales[[
    "product_name",
```

```
    "quantity",
    "unit_price_x",
    "discount_percent",
    "sale_amount"
]].head(10))

print("Ümumi satış:", sales["sale_amount"].sum())
```

	product_name	quantity	unit_price_x	discount_percent	sale_amount
0	Hoodie	1	89	0	89.00
1	Hoodie	1	89	20	71.20
2	Elektrik çaydanı	3	69	10	186.30
3	Bluetooth qulaqcıq	2	59	0	118.00
4	Blender	1	89	0	89.00
5	Simsiz mouse	1	39	20	31.20
6	Saç maskası	1	27	15	22.95
7	Simsiz mouse	1	39	5	37.05
8	Makiyaj dəsti	2	68	0	136.00
9	Üz kremi	1	34	10	30.60

```
Ümumi satış: 93692.4
```

```
sales["order_date"] = pd.to_datetime(sales["order_date"])
sales["month"] = sales["order_date"].dt.to_period("M").astype(str)

monthly_sales = sales.groupby("month")["sale_amount"].sum()
display(monthly_sales)
```

month	sale_amount
2025-01	4405.10
2025-02	3671.15
2025-03	6965.40
2025-04	5239.90
2025-05	6264.75
2025-06	6496.45
2025-07	3341.40
2025-08	4768.75
2025-09	3790.00
2025-10	5165.75
2025-11	7169.60
2025-12	5465.70
2026-01	4237.95
2026-02	4998.15
2026-03	5087.15
2026-04	6643.65
2026-05	5191.65
2026-06	4789.90

dtype: float64

```
display(
    sales.groupby("product_name")["sale_amount"]
    .sum()
    .sort_values(ascending=False)
)
```

product_name	sale_amount
Smart saat	9449.25
Don	8794.50
Blender	7039.90
Hoodie	6652.75
Ətir	6412.50
Elektrik çaydanı	6106.50
Powerbank	5712.80
Cins şalvar	5265.35
Makiyaj dəsti	5123.80
Dumbbell set	4887.75
İdman çantası	4042.50
Bluetooth qulaqcıq	3634.40
Simsiz mouse	3492.45
Qadın köynəyi	3264.75
Termos	2901.50
Yoga matı	2741.70
Üz kremi	2675.80
Stol lampası	2114.70
Saç maskası	2097.90
Fitness rezini	1281.60

dtype: float64

```
display(
    sales.groupby("category")["sale_amount"]
    .sum()
    .sort_values(ascending=False)
)
```

category	sale_amount
Geyim	23977.35
Elektronika	22288.90
Ev və Mətbəx	18162.60
Gözəllik	16310.00
İdman	12953.55

dtype: float64

```
display(
    sales.groupby("city")["sale_amount"]
    .sum()
    .sort_values(ascending=False)
)
```

city	sale_amount
Bakı	46296.25
Sumqayıt	14670.60
Şəki	10168.90
Lənkəran	9690.25
Gəncə	8485.75
Mingəçevir	4380.65

dtype: float64

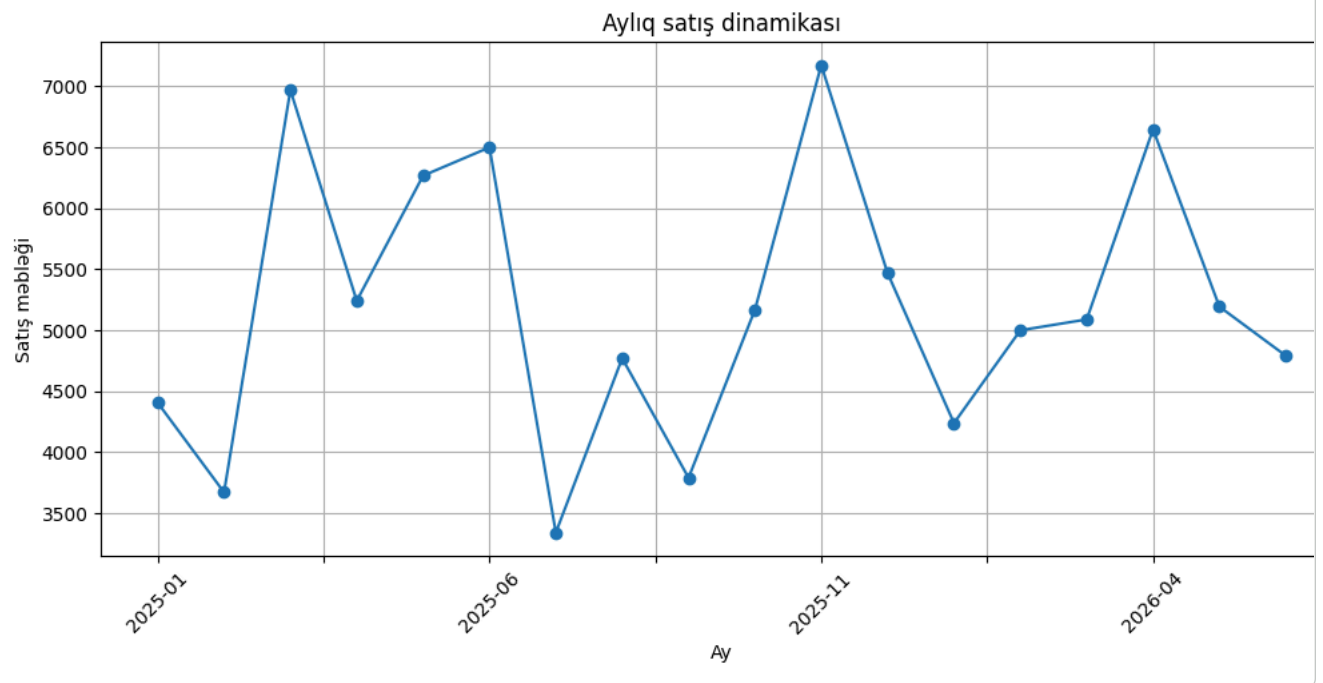
```
import matplotlib.pyplot as plt
```

```
sales["order_date"] = pd.to_datetime(sales["order_date"])
sales["month"] = sales["order_date"].dt.to_period("M").astype(str)

monthly_sales = sales.groupby("month")["sale_amount"].sum()

monthly_sales.plot(
    kind="line",
    marker="o",
    figsize=(12,5),
    title="Aylıq satış dinamikası"
)

plt.xlabel("Ay")
plt.ylabel("Satış məbləği")
plt.xticks(rotation=45)
plt.grid()
plt.show()
```

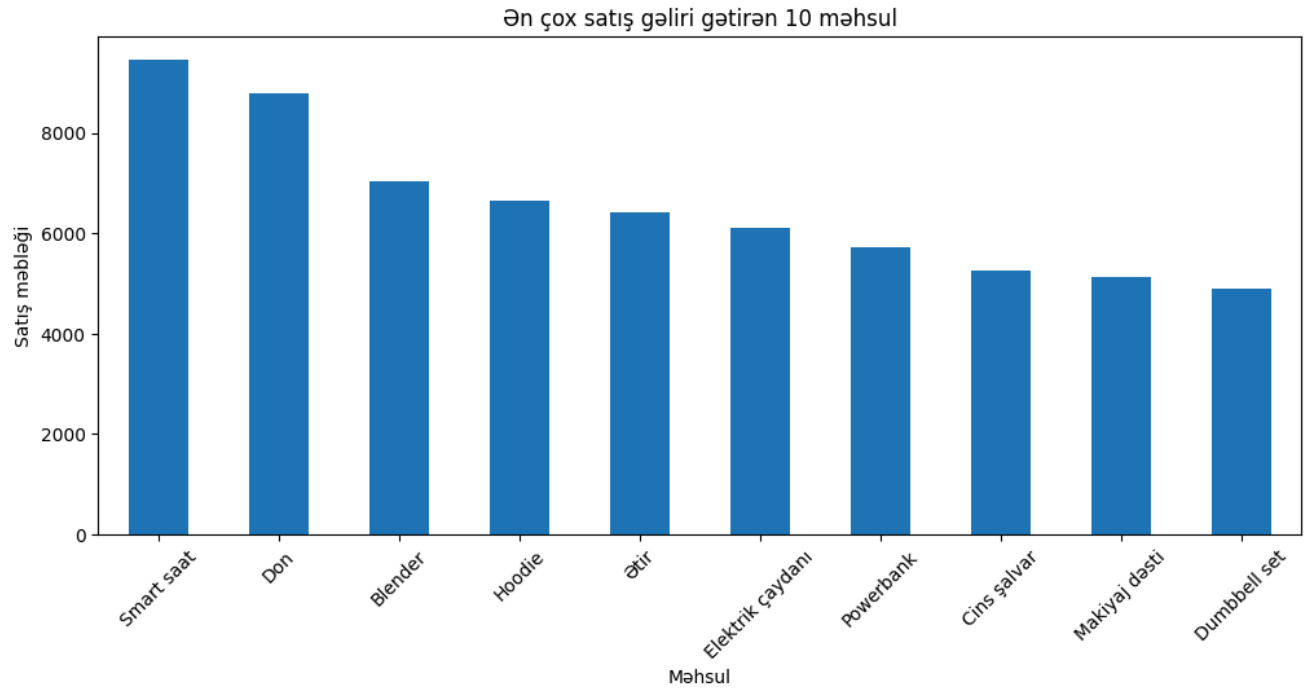



```
product_sales = (
    sales.groupby("product_name")["sale_amount"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

product_sales.plot(
    kind="bar",
    figsize=(12,5),
    title="Ən çox satış gəliri gətirən 10 məhsul"
```

```
)

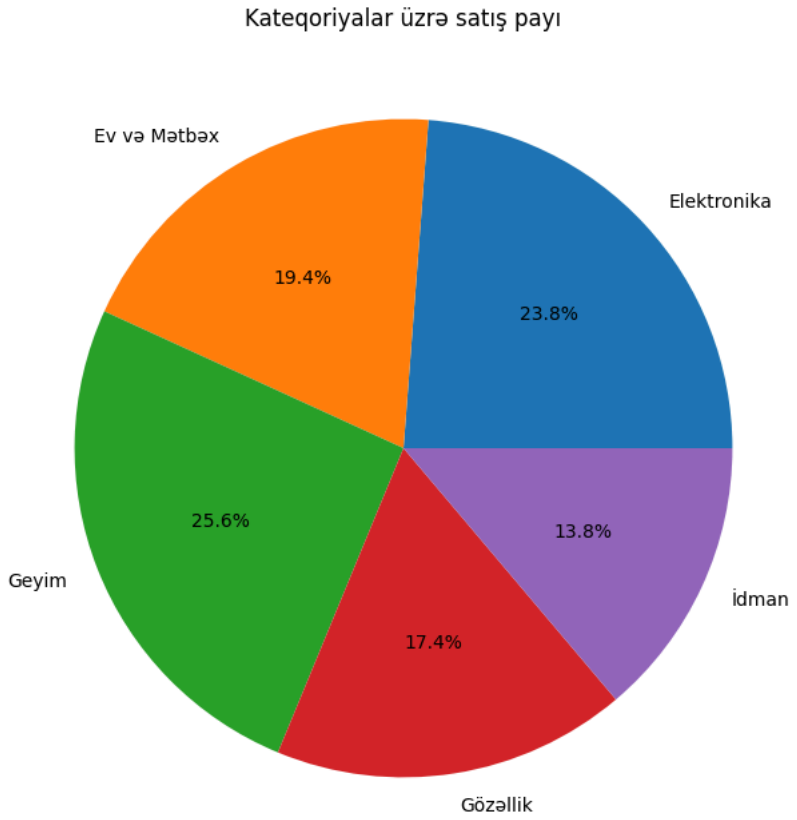
plt.xlabel("Məhsul")
plt.ylabel("Satış məbləği")
plt.xticks(rotation=45)
plt.show()
```




```
category_sales = sales.groupby("category")["sale_amount"].sum()

category_sales.plot(
    kind="pie",
    autopct="%1.1f%%",
    figsize=(8,8),
    title="Kateqoriyalar üzrə satış payı"
)

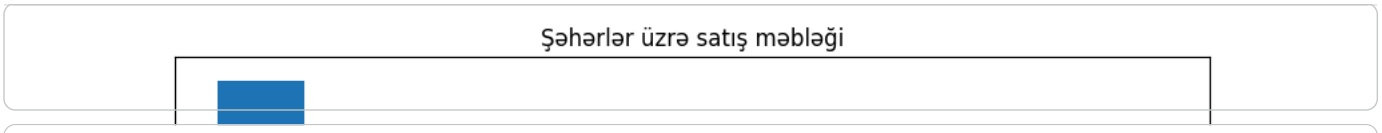
plt.ylabel("")
plt.show()
```




```
city_sales = (
    sales.groupby("city")["sale_amount"]
    .sum()
    .sort_values(ascending=False)
)

city_sales.plot(
    kind="bar",
    figsize=(10,5),
    title="Şəhərlər üzrə satış məbləği"
)

plt.xlabel("Şəhər")
plt.ylabel("Satış məbləği")
plt.xticks(rotation=45)
plt.show()
```




```
##Maraqlı nəticələri çıxaran Python kodu
# 1. Ən çox satış olan ay
best_month = monthly_sales.idxmax()
best_month_value = monthly_sales.max()

# 2. Ən çox satılan məhsul
best_product = product_sales.idxmax()
best_product_value = product_sales.max()

# 3. Ən çox satış gəliri olan kateqoriya
best_category = category_sales.idxmax()
best_category_value = category_sales.max()

# 4. Ən cox satıs olan səhər
```